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Reducing dietary crude protein: effects on digestibility, N balance, and blood metabolites in late-lactation Holstein cows

M. G. Erickson,¹ T. Barros,¹ M. J. Aguerre,² J. J. Olmos Colmenero,³ S. J. Bertics,¹ and M. A. Wattiaux^{1*}

¹Department of Animal & Dairy Sciences, University of Wisconsin-Madison, Madison 53706

²Department of Animal and Veterinary Sciences, Clemson University, Clemson, 29634

³Departamento de Ciencias Pecuarias y Agrícolas, Centro Universitario de Los Altos de la Universidad de Guadalajara, Tapatitlán, Jalisco, México, 47600

ABSTRACT

Our objectives were to determine the effects of reducing dietary crude protein (CP) concentration on nutrient digestibility, rumen function, N balance, and serum AA concentration for dairy cows in late lactation. At the initiation of the experimental period, we stratified Holstein cows ($n = 128$; mean $\pm$ standard deviation 224 ± 54 d in milk) by parity and days pregnant (86 ± 25 d) and assigned them to 1 of 16 pens. For 3 wk, all cows received a covariate diet containing 16.9% CP [dry matter (DM) basis]. For the subsequent 12 wk, we assigned pens to 1 of 4 treatments containing 16.2, 14.4, 13.4, or 11.9% CP (DM basis) in a randomized complete block design. Diets were fed as a total mixed ration once daily. To reduce dietary CP, we replaced soybean meal with soybean hulls in the concentrate mix (DM basis). Diet evaluations suggested that several EAA, especially His, limited productivity as dietary CP declined. Digestibility of DM and CP decreased linearly with dietary CP reduction. Digestibility of neutral detergent fiber and potentially digestible neutral detergent fiber tended to respond in a quadratic pattern with the greatest digestibility at intermediate treatments. The reduction in dietary CP did not affect ruminal pH, but ruminal ammonia-N and branched-chain VFA concentrations declined linearly. The concentration of milk urea-N and plasma urea-N, secretion of milk N, and excretions of fecal N, urinary N, urinary urea-N, and unaccounted N decreased linearly with the reduction in dietary CP concentration. Urinary N expressed as a percentage of N intake was unaffected by dietary CP. Serum concentrations of total essential AA and non-essential AA were unaffected by dietary CP concentration. However, the ratio of essential to non-essential AA decreased with decreasing dietary CP. Serum 3-methylhistidine concentration increased linearly with decreasing dietary

CP concentration, indicating greater skeletal muscle breakdown. Although our trial confirmed that reducing dietary CP decreased absolute excretion of urinary N, diet evaluations suggested that milk protein production decreased as certain essential AA became increasingly limited. Thus, reduced-CP diets have the potential to lessen reactive-N outputs of late lactation cows, but more research is needed to design diets that minimize deleterious effects on productivity.

Keywords: **rumen, urea, ammonia, amino acid, 3-methylhistidine**

INTRODUCTION

Reducing environmentally-reactive N losses in manure is a priority to enhance the dairy industry's environmental sustainability (Steinfeld et al., 2006). Reactive N in manure is associated with environmental impacts including greenhouse gas emissions, air pollution, and N runoff and leaching (Lee et al., 2011; Chadwick et al., 2018). Compared with fecal N, managing urinary N output is particularly important because urinary N is comprised primarily of reactive N compounds such as urea (Spek et al., 2013b). Because reducing N intake is a reliable means to decrease urinary N excretion, there is growing interest in reducing dietary CP levels (Dijkstra et al., 2011; Mutsvangwa et al., 2016). For example, simulation studies have shown that groupings that allow the feeding of reduced dietary CP in late lactation may reduce N excretion, improve N efficiency, and lessen feed costs (Allen, 2009; Barrientos-Blanco et al., 2020, 2022). However, existing research on reduced-CP diets has raised questions about the optimal conditions for reducing dietary CP on dairy farms.

Much of the existing research on reduced-CP diets involved cows in early- to mid-lactation when milk protein synthesis can drive substantial metabolizable protein demands (Broderick, 2003; Colmenero and Broderick, 2006). Conversely, reduced-CP diets may be more practical in late lactation. In a full-lactation feeding trial by Law et al. (2009), reducing dietary CP from

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*Corresponding author: wattiaux@wisc.edu

17.3 to 14.4% of DM after d 151 of lactation had no effects on milk yield, milk protein yield, or BW change and improved N use efficiency (NUE) compared with maintaining 17.3% dietary CP. Recently, an 8 wk trial showed that greater CP level and more advanced stage of lactation independently reduced NUE, which confirmed that excess dietary CP in relation to decreasing animal requirements may lead to increased manure N excretion as lactation advances (Letelier et al., 2022b). Still, reductions in dietary CP that achieve greater NUE can reduce milk protein production and impose on body protein reserves (Broderick et al., 2008; Liu et al., 2021). As several recent studies have contended, the potential undesirable effects of low dietary CP may be exacerbated when the dietary supply of digestible EAA is low relative to predicted animal requirements (Omphalius et al., 2019; Räisänen et al., 2021). The combined effects of reduced dietary AA supply and counterbalancing physiological mechanisms (e.g., lower milk protein secretion, increased muscle protein degradation) may be reflected in blood AA concentrations (Letelier et al., 2022a).

In this study we characterized the response of late lactation cows to reductions in dietary CP for a 12-week period. We examined indicators of N digestion and metabolism across CP levels predicted to impose slight (16.2% CP) to severe (11.9% CP) deficiencies in MP and metabolizable AA (NASEM, 2021). We hypothesized that indicators of ruminal fermentation and N metabolism would respond in a quadratic manner, with minimal response to dietary CP content until more severe deficiencies (11.9 and 13.4% CP). We expected to see indications of greater muscle protein mobilization as dietary CP decreased, resulting in limited effects of dietary CP on blood AA concentrations. Overall, we aimed to determine the pattern of responses to incremental reductions in dietary CP content in late lactation to quantify the physiological and productivity trade-offs potentially associated with reducing reactive-N excretion.

MATERIALS AND METHODS

The Institutional Animal Care and Use Committee of the College of Agricultural and Life Sciences of the University of Wisconsin-Madison approved the experimental protocol for this study, which was conducted from September to December 2013 at the University of Wisconsin-Madison Emmons-Blaine experimental station (Arlington, WI). This study was part of a larger experiment that evaluated the effects of decreasing dietary CP concentration in late-lactation dairy cows over a 12-wk period. The present study focuses on the rumen environment, total-tract apparent digestibility,

N partitioning, and serum AA concentrations. Data on animal productivity and efficiency of N utilization has been reported by Barros et al. (2017).

Cows, Dietary Treatments, and Experimental Design

Experimental procedures were detailed in Barros et al. (2017). Briefly, this study used 128 Holstein cows in late lactation at 224 ± 54 DIM (mean $\pm$ SD) at the initiation of the experimental period. Cows were stratified by parity (2.5 ± 1.3 lactations) and days pregnant (86 ± 25 d) and assigned to 1 of 16 pens (experimental units), each including 2 primiparous and 6 multiparous cows. The study began with a 3-wk covariate period during which all cows were fed a diet containing 16.9% CP (DM basis). For the subsequent 12 weeks (experimental period), pens were blocked by 1 of 2 barn wings and assigned to 1 of 4 dietary treatments containing 16.2, 14.4, 13.4, and 11.9% CP (DM basis) in a randomized complete block design. Diet DM comprised the forages corn silage and alfalfa silage (65.9 to 66.3%), high moisture corn (12.9 to 13.3%), and a concentrate mix (20.4 to 22.0%). The CP reduction in the diet was achieved by replacing soybean meal with soy hulls in the concentrate mix, which was prepared at the University of Wisconsin Feed Mill (Arlington, WI). Diets were fed as a TMR once daily at 0700 h allowing *ad libitum* feeding targeting 5 to 10% refusal. As discussed in Barros et al. (2017), all experimental diets were lower CP than the covariate, and all experimental diets were formulated to limit MP based on NRC (2001). Subsequent evaluation of diets and animal performance using an updated model (NASEM, 2021; Appendix Table A1) showed that EAA efficiencies exceeded target efficiencies for His, Met, and Leu for all levels of dietary CP (Appendix Table A2). At lower levels of dietary CP, nearly all EAA efficiencies exceeded targets. This suggested that EAA supplies, in addition to the dietary CP manipulation, may have affected outcomes.

Feed and Milk Sampling

Feed and milk samples were collected weekly as described by Barros et al. (2017). Data from the covariate period and wk 3, 7, and 11 of the experiment are presented in the current study. Samples of feed ingredients (corn silage, alfalfa silage, high moisture corn, concentrate mixes), TMR, and refusals were processed and analyzed as previously described (Barros et al., 2017). Dry matter and N intake were determined on a pen basis using weekly absolute DM percentages and daily weights of as-fed TMR offered and refused, which were obtained from the Feed Supervisor Software (Supervisor Systems, Dresser, WI). Milking procedures, sampling,

and analysis were described by Barros et al. (2017). In summary, samples were taken at 2 consecutive milkings, preserved with 2-bromo-2-nitropropan-1,3-diol, and analyzed via infrared spectrometry. Each cow's milk yield was measured at every milking throughout the experiment.

Fecal and Urine Sampling and Analysis

Spot urine samples were collected by manual stimulation of the peri-vulvar region of 4 cows per pen (3 of 6 multiparous and 1 of 2 primiparous cows were randomly selected from each pen at each time point) and composited by equal volumes per animal within each pen and time point, 6 h before feeding, on 1 d during wk 3, 7 and 11 of the treatment period. This protocol was repeated 6 h after feeding. From each urine composite, a 15 mL sub-sample was acidified by dilution with 60 mL of 0.072 N H₂SO₄ and stored in a specimen cup at −20°C until further analysis. Urine urea-N and creatinine were analyzed on a flow injection analyzer using colorimetric assays (Broderick and Clayton, 1997; Lachat Quik-Chem 8000 FIA, Lachat Instruments, Milwaukee, WI). Urine output was estimated using creatinine as a marker and assuming an excretion of 29 mg/kg of BW per day (Valadares et al., 1999).

For wk 3, 7, and 11 of the experimental period, grab samples of feces were collected from 6 cows per pen (at each time point, 5 of 6 multiparous and 1 of 2 primiparous cows were randomly selected from each pen) and composited by pen by equal volumes. Each sampling week, fecal samples were collected at staggered time points of 1200, 2400, 0600, and 1800 h over 2 d to cover a 24-h feeding cycle and stored at −20°C until further analysis. Composite samples by pen and time point were dried in a forced-air oven (60°C, 48 h) and ground in a Wiley mill (model 4; A. H. Thomas Co. Philadelphia, PA) through a 1-mm screen. Fecal samples were also analyzed for NDF, starch, and ash according to the methods described for feeds by Barros et al. (2017). Fecal and urine samples were analyzed for total N by a combustion assay (Leco FP-2000 Nitrogen Analyzer, Leco Instruments Inc., St. Joseph, MI). Unaccounted N was calculated as the difference between N intake and the sum of N recovered in feces, urine, and milk.

Digestibility

Fecal output and total-tract apparent nutrient digestibility were calculated using 288-h indigestible NDF (iNDF) as an internal marker. The concentration of iNDF was determined in feed components, refusals, and feces according to Krizsan and Huhtanen (2013) with the following modifications: One-gram samples

were weighed into F57 fiber bags (Ankom Technology, Macedon, NY) in quadruplicate; Samples were incubated in situ for 288 h in 2 ruminally-cannulated Holstein multiparous lactating dairy cows fed a TMR containing (DM basis) alfalfa silage (44.5%), corn silage (26.8%), alfalfa hay (10.7%), wheat straw (6.5%), and concentrate mixture (11.5%). After removal from the rumen, bags were washed according to Cherney et al. (1990) dried at 55°C in a forced-air oven for 48 h, and analyzed for NDF as described by Barros et al. (2017). The amount of potentially-digestible NDF (pdNDF) was calculated as the total NDF minus indigestible NDF.

Rumen Sampling and Analysis

Four cows per pen (3 multiparous and 1 primiparous) were randomly selected once at the beginning of the trial for rumenocentesis throughout the trial (Nordlund and Garrett, 1994). For each cow, a rumen fluid sample was collected once, approximately 4 h after feeding on wk 3 of the covariate period and wk 4, 8, and 12 of the experimental period. Within 3 min of collection, pH was determined using 2 pH meters (Laqua Twin pH-meter model B-713; Spectrum Technologies Inc., Plainfield, IL). Four 1-mL aliquots were pipetted into microcentrifuge tubes before freezing at −20°C. Two aliquots were preserved with 20 µL of 50% trichloroacetic acid for analysis of NH₃-N, and 2 were preserved with 20 µL of 50% H₂SO₄ solution for analysis of VFA concentrations. A colorimetric assay was used to determine rumen NH₃-N concentration (Chaney and Marbach, 1962). Rumen VFA concentrations were measured using a Clarus 500 gas chromatograph (PerkinElmer, Norwalk, CT) using a 4% Carbowax 20 M on 80/120 mesh Carbowax-B-DA, 1.8 mm × 2 mm column (Supelco Inc., Bellefonte, PA) with an oven temperature of 160°C and an N flow rate of 24 mL/min.

Blood Sampling and Analysis

On d 15 of the covariate period at 1100 h a blood sample was collected from the coccygeal vessels of all cows using evacuated serum tubes (BD Vacutainer, Becton Dickinson, Franklin Lakes, NJ). On wk 4 and wk 12, the same 4 cows per pen selected for rumenocentesis were also subject to blood sampling once approximately 5 h after feeding using evacuated serum tubes. Serum was obtained by centrifugation at 1000 *g* for 15 min at room temperature and stored at −20°C until further analysis. A colorimetric assay was used to determine BUN concentration in individual samples (Chaney and Marbach, 1962). Serum samples were composited (by equal volume) by pen and sent to

Heartland Laboratories (Ames, IA) for analysis of AA, including 3-MH. Serum AA analysis was carried out by liquid chromatography-electrospray-tandem mass spectrometry (LC-ESI-MS/MS) using an Agilent 1200/6460 triple quadrupole LC/MS system (Santa Clara, CA). Standards and 25 μ L of serum were processed using an EZ:faastTM (Phenomenex, Torrance, CA) analysis kit for physiological AA by LC-MS (Badawy, 2012). Briefly, the procedure consisted of solid phase extraction followed by derivatization (propyl chloroformate) of the AA and liquid-liquid extraction. Chromatographic separation of the derivatized AA was conducted on an EZ:faast AA analysis-mass spectrometry column (250 $\times$ 2.0 mm i.d., 4 μ m). Ammonium formate in water (10 mM) and ammonium formate in methanol (10 mM) served as the eluents. A linear gradient was used, spanning 68–83% methanol for 0–13 min. We used the Mass Hunter acquisition software (version B.4.01, Agilent, Santa Clara, CA) to optimize procedures, collect data, and quantitate AA using linear standard lines.

Statistical Analysis

Five cows were removed from the study and not replaced. Removals included 4 cows in the 13.4% CP treatment in wk 1 (pen 5), wk 2 (pen 12), wk 9 (pen 14), and wk 12 (pen 5) due to mastitis, pneumonia, dry-off, and dry-off, respectively. An additional cow was removed from the 11.9% CP treatment in wk 8 (pen 15) due to lameness. We analyzed data with the MIXED procedure of SAS version 9.3 (2012) as a completely randomized block design. Pen was the experimental unit with measurements repeated over week as shown:

$$Y_{ijkl} = \mu + C + T_i + B_k + P_l(B_k \times T_i) + W_j + (W_j \times T_i) + e_{ijkl},$$

In this model, Y_{ijkl} is the response variable; μ is the overall mean; C is the fixed effect of the covariate measurement; T_i is the fixed effect of treatment ($i = 1-4$); B_k is the random effect of barn wing ($k = 1-2$); $P_l(B_k \times T_i)$ is the random effect of pen ($l = 1$ to 16) nested within barn wing and treatment combination; W_j is the fixed effect of week ($j = 1$ to 3); $(W_j \times T_i)$ is the week by treatment interaction, and e_{ijkl} is the random residual error. Digestibility and N excretion data were analyzed using the above model without covariate. We selected covariance structures based on Akaike information criteria: using a spatial power structure for serum AA, and a first-order autoregressive structure for other variables. Degrees of freedom were calculated using the Kenward-Roger option (Schaalje et al., 2001). Orthogonal polynomial contrasts were designed using

the unequally-spaced dietary CP concentrations to determine the linear and quadratic effects of treatments on response variables. In the presence of a significant treatment $\times$ week interaction effect, the means were plotted and the source of the interaction was explored using the SLICE option (SAS, 2012). Statistical significance and trends were considered at $P < 0.05$, and $P \geq 0.05$ to $P < 0.10$, respectively.

RESULTS AND DISCUSSION

This study examined the response of late lactation cows to reductions in dietary CP (16.2 to 11.9% of DM) over a 12-week period. Using pens with primiparous and multiparous late-lactation cows, we aimed to characterize the pattern of responses to incremental reductions in dietary CP to examine the trade-offs between productivity and reactive N excretion. In general, our results suggested linear and quadratic but not cubic responses. Importantly, our trial focused on pen-level responses and this may mask important animal-level variation in responses. Additionally, more research is needed to determine if the tendencies observed in our trial reflect true, small magnitude effects that were not statistically significant at the sample size used in our trial.

Dietary Composition, Animal Performance, and Milk Composition

Table 1 shows diet composition on average across wk 3, 7, and 11, which did not differ substantially from the study-wide averages presented by Barros et al. (2017). As designed, replacing soybean meal with soybean hulls decreased CP concentration (16.2, 14.4, 13.4, and 11.9% of DM), increased non-forage NDF concentration (4.8, 6.8, 9.4, and 10.4% of DM), and slightly decreased NRC-predicted NE_L concentration (1.57, 1.56, 1.53, and 1.52 Mcal/kg of DM) while starch concentration remained relatively constant (22.1 to 22.4% of DM). Diets were formulated using NRC (2001) so that the percentages of His (2.1%), Met (1.8%), and Lys (6.7%) in MP were uniform across dietary CP concentrations. Retrospective evaluation of the diets using an updated model (NASEM, 2021; Appendix Table A1) suggested that predicted EAA efficiencies exceeded target efficiencies for His, Met, and Leu when dietary CP was 16.2% of DM (Appendix Table A2). At lower dietary CP concentrations, nearly all predicted EAA efficiencies exceeded targets. Cow performance averaged across wk 3, 7, and 11 and reported in Table 2 did not differ substantially from the study-wide averages reported by Barros et al. (2017). Reducing dietary CP caused linear reductions in fat-and-protein-corrected milk (FPCM),

Table 1. Ingredient and chemical composition of covariate diet in wk 2 and 3 of the covariate period and dietary treatments averaged for wk 3, 7, and 11 of the experimental period

	Covariate	Dietary CP Treatments (DM basis)			
		16.2	14.4	13.4	11.9
Ingredient, % of DM					
Corn silage	31.3	33.9	33.9	33.8	34.0
Alfalfa silage	33.6	32.0	32.4	32.2	32.2
High moisture corn	13.1	12.9	13.3	13.0	13.3
Concentrate mix					
Soybean meal, 48%	12.29	11.84	7.60	3.91	0.00
Soy hull pellets	0.00	0.00	3.78	7.82	11.45
Beet pulp	5.43	5.24	5.04	5.18	5.06
Molasses	2.27	2.19	2.11	2.17	2.12
Calcium carbonate	0.73	0.70	0.68	0.70	0.68
TMV premix ¹	0.44	0.41	0.41	0.42	0.41
Monosodium phosphate (1H ₂ O)	0.40	0.38	0.37	0.38	0.37
Magnesium Oxide	0.32	0.32	0.30	0.30	0.30
Sodium Chloride	0.12	0.12	0.12	0.12	0.12
Nutrient ²					
DM (% as fed)	53.9	53.2	52.8	52.8	52.9
CP	16.9	16.2	14.4	13.4	11.9
Non-forage NDF	4.5	4.8	6.8	9.4	10.4
Forage NDF	24.7	26.7	26.9	26.8	26.9
NDF	29.2	31.5	33.7	36.2	37.3
Ether extract	3.64	3.59	3.57	3.61	3.61
Ash	7.15	7.66	7.72	7.39	7.20
NFC ³	45.3	43.3	43.0	41.7	42.4
Starch	21.2	22.1	22.4	22.1	22.4
NDIN ⁴	0.35	0.37	0.37	0.38	0.39
NE _L ⁵ (Mcal/kg of DM)	1.58	1.57	1.56	1.53	1.52

¹Trace Mineral and Vitamin premix (DM basis): 52.3% of Cl, 34.7% of Na; 0.5% of Ca; 2.11% of Zn; 1.47% of Mn; 0.50% of Cu; 0.09% of S; 479 mg/kg I; 91 mg/kg of Se; 80 mg/kg of Co; 2,103,287 IU/kg of Vitamin A; 421,108 IU/kg of Vitamin D; 8,918 IU/kg of Vitamin E.

²Nutrients expressed as % of DM unless stated otherwise.

³NFC = 100 – (CP % + (NDF % – NDICP %) + ether extract % + ash %).

⁴Neutral detergent insoluble nitrogen.

⁵Net energy of lactation estimated using NRC (2001) with DMI, BW, milk yield, and milk composition observed during the trial.

milk true protein concentration, and milk component yields (true protein, fat, lactose). Barros et al. (2017) found that yields of milk and several milk components responded quadratically to dietary CP decreasing productivity gains at greater dietary CP. For the subset of weeks considered in the present study, yields of milk and lactose tended to respond in a quadratic pattern similar to Barros et al. (2017).

Intake and Apparent Digestibility

Intake of DM, CP, and starch (kg/d) decreased linearly with the reduction of the dietary CP concentration (Table 3). The decrease in CP intake was due to both decreasing DMI and the reduction of dietary CP concentration. Based predominantly on the dietary substitution of soybean meal with soybean hulls, intake of NDF and pdNDF increased linearly with reducing dietary CP. Additionally, there was a quadratic effect where NDF and pdNDF intake were the greatest with

the 13.4% CP treatment. The effects of CP on DMI have been inconsistent in the literature. Researchers have reported insignificant differences in DMI with moderate changes in dietary CP in several prior studies: Broderick et al. (2015) with 15.0 vs. 17.0% CP, Oh et al. (2019) with 15.5 vs. 16.5% CP, and Tebbe and Weiss (2020) with 14.1 vs. 16.2% CP. In contrast, Letelier et al. (2022b) found linear, quadratic, and cubic effects of dietary CP level on DMI with a 2.8 kg/d reduction for a dietary CP concentration of 13.6 compared with 16.7%. In a meta-analysis, increased feed intake and milk production was reported with increasing dietary protein intake (Huhtanen and Hetta, 2012). In our study, the reduction of dietary CP and increase in fiber concentrations resulted in numerically equivalent DMI for 16.2 and 13.4% CP, which possibly influenced digestibility results.

Despite decreasing DMI, the apparent digestibility of DM and CP declined linearly with the reduction in dietary CP. Decreased CP digestibility appeared

Table 2. Effect of dietary CP content on daily cow performance averaged across wk 3, 7, and 11 for an experiment with n = 16 pens and n = 128 cows.¹

Item	Dietary CP Treatments (DM basis)					P-value ²		
	16.2	14.4	13.4	11.9	SEM	L	Q	Trt × Wk
DM intake, kg/d	24.4	23.9	24.3	23.0	0.27	0.01	0.20	0.55
Milk yield, kg/d	31.3	30.0	27.7	24.2	0.86	<0.01	0.07	0.34
FPCM ³ , kg/d	32.4	30.9	29.3	25.5	0.88	<0.01	0.12	0.24
Milk composition								
True protein, %	3.59	3.47	3.51	3.33	0.03	<0.01	0.19	0.03
Fat, %	4.25	4.08	4.38	4.37	0.10	0.25	0.36	0.50
Lactose, %	4.70	4.69	4.67	4.68	0.03	0.48	0.85	0.10
Milk components yield								
True protein, kg/d	1.13	1.03	0.97	0.80	0.03	<0.01	0.25	0.44
Fat, kg/d	1.30	1.22	1.21	1.06	0.04	<0.01	0.32	0.34
Lactose, kg/d	1.47	1.42	1.31	1.11	0.06	<0.01	0.08	0.42

¹Covariate-adjusted least squares means.²P-values for linear (L), quadratic (Q), treatment (Trt), or treatment × week (Trt × Wk) effect of dietary CP. Week effect was significant ($P \leq 0.01$) for all response variables.³FPCM (fat-and-protein-corrected milk) = milk (kg/d) * [0.1226 * fat (%) + 0.0776 * protein (%) + 0.2534] (IDF, 2015).

to partially explain the reduction in DM digestibility. Lesser dietary CP is expected to decrease apparent CP digestibility because metabolic fecal N is diluted to a progressively smaller extent by undigested dietary CP (Holter et al., 1982). A similar dilution effect was supported by our study and other recent work (Giallongo et al., 2015; Tebbe and Weiss, 2020; Räisänen et al., 2022). In contrast to CP, the digestibility of NDF and pdNDF is not confounded by endogenous sources. In our trial, NDF digestibility tended to respond qua-

dratically to CP and was greatest for the 2 intermediate treatments. Compared with forage NDF sources, soybean hulls may have greater NDF digestibility due to lower lignin content and greater surface area (Negrão et al., 2020). Although starch content was similar across treatments, the 16.2% CP treatment resulted in a greater amount of digested starch, which may explain the modest depression in NDF digestibility for this treatment (De Souza et al., 2018). Greater dietary CP (up to 22% of diet DM) has been associated with

Table 3. Effect of dietary CP concentration on nutrient intake and digestibility for n = 16 pens of late-lactation dairy cows with samples and measurements for 2 d on wk 3, 7, and 11.¹

Item	Dietary CP Treatments (DM basis)					P-value ²		
	16.2	14.4	13.4	11.9	SEM	L	Q	Trt*wk
Intake, kg/d								
DM	24.6	23.9	24.6	22.7	0.37	0.01	0.14	0.70
CP	4.00	3.44	3.30	2.71	0.06	<0.01	0.79	0.03
NDF	7.98	8.32	9.21	8.70	0.13	<0.01	0.02	0.41
pdNDF ³	5.21	5.57	6.38	6.07	0.09	<0.01	0.02	0.18
Starch	5.58	5.44	5.50	5.18	0.09	<0.01	0.28	0.94
Digestibility, %								
DM	68.7	68.3	66.9	65.2	0.63	<0.01	0.15	0.33
CP	67.2	64.3	61.2	56.6	0.85	<0.01	0.12	0.18
NDF	48.4	50.7	50.9	48.8	1.14	0.78	0.05	0.18
pdNDF	74.1	75.8	73.5	70.1	1.67	0.05	0.06	0.27
Starch	97.1	97.1	97.0	96.9	0.12	0.12	0.53	0.08
Mass digested, kg/d								
DM	16.9	16.3	16.4	14.8	0.29	<0.01	0.10	0.29
CP	2.69	2.21	2.02	1.53	0.04	<0.01	0.22	0.06
NDF	3.86	4.21	4.68	4.25	0.09	<0.01	<0.01	0.17
Starch	5.42	5.28	5.33	5.02	0.09	0.01	0.27	0.95

¹Covariate-adjusted least squares means.²P-values for linear (L), quadratic (Q), or treatment × week (Trt × Wk) effect of dietary CP. Week effect was significant ($P \leq 0.01$) for all response variables.³Potentially digestible NDF; pdNDF = NDF – Indigestible NDF.

small improvements in NDF and pdNDF digestibility in meta-analysis (Huhtanen and Hristov, 2009) and individual trials (e.g., Broderick, 2003), although this was not the case with our study.

Rumen Environment

Table 4 presents treatment effects on the rumen environment. Ruminal pH was unaffected by the reduction in dietary CP concentration. However, ruminal $\text{NH}_3\text{-N}$ concentration decreased linearly by 0.82 ± 0.16 mg/dL per percentage unit of dietary CP. This was similar to the 0.87 ± 0.15 mg/dL reduction reported by Colmenero and Broderick (2006) for each percentage unit CP reduction from 19.4 to 13.5%. The decrease in $\text{NH}_3\text{-N}$ did not seem to alter microbial fermentation activity in our experiment, because concentrations of major VFA were unaffected by reduced CP. However, decreasing dietary CP concentration linearly reduced the total concentration of branched-chain VFA (BCVFA; isobutyric, isovaleric, and 2-methyl butyric acids). Specifically, in our trial, isobutyrate and 2-methylbutyrate tended to decrease linearly with the reduction of dietary CP concentration, similar to Agle et al. (2010). Isobutyric, isovaleric, and 2-methyl butyric acids are produced in the rumen upon deamination and decarboxylation of the branched-chain AA Val, Leu, and Iso, respectively (Roman-Garcia et al., 2021). Reduced concentrations of ruminal $\text{NH}_3\text{-N}$ and BCVFA could suggest lesser ruminal protein degradation (Van Soest, 1994). Differences in the AA composition of RDP may also drive differences in BCVFA, as suggested by studies comparing supplemental protein sources (Broderick et al., 2015). In the present study, there was no further reduction in BCVFA concentration between the 2 lower dietary

CP concentration treatments, although $\text{NH}_3\text{-N}$ declined from 7.1 to 6.3 mg/dL. Decreased DMI in the 11.9% CP treatment may have also decreased ruminal digesta pool size, potentially contributing to maintaining BCVFA and $\text{NH}_3\text{-N}$ concentrations by altering ruminal digestion kinetics. Because BCVFA may stimulate microbial fibrolytic activity (Misra and Thakur, 2001; Broderick, 2003; Detmann et al., 2009), reducing dietary CP by lowering RDP may lessen ruminal fiber digestibility. In our trial, reducing dietary RDP from 11.1 to 8.4% of DM (NRC, 2001) decreased ruminal BCVFA and $\text{NH}_3\text{-N}$ concentrations linearly, yet digestibility of NDF and pdNDF responded quadratically with maxima at intermediate levels of dietary CP.

Blood, Milk, and Urinary Urea-N and N balance

Results for variables related to blood, milk, and urinary urea-N and N balance are presented in Table 5. As intake N decreased in our trial (46 g per percentage unit CP), the amounts of fecal and urine N decreased linearly (4 g and 16 g per percentage unit CP, respectively) and milk N declined linearly (11 g per percentage unit CP). Correspondingly, with decreasing dietary CP, fecal N represented a greater proportion of N intake ($P < 0.01$), and urinary N accounted for a progressively lesser fraction of intake N ($P = 0.23$). Despite reducing the amount of manure N excreted, reducing dietary CP did not improve NUE. Thus, our trial demonstrated a case when reducing dietary CP concentration lessened the overall amount of N excreted but did not reduce N excretion per unit of milk.

Unaccounted N decreased linearly with decreasing dietary CP in our trial (Table 5). Meta-analyses have suggested unaccounted N is generally greater than zero

Table 4. Effect of dietary CP content on rumen parameters on wk 4, 8, and 12 for $n = 16$ pens of late lactation cows.¹

Item	Dietary CP% Treatments (DM basis)				SEM	<i>P</i> -value ²	
	16.2	14.4	13.4	11.9		L	Q
pH	6.14	6.21	6.10	6.21	0.06	0.63	0.80
$\text{NH}_3\text{-N}$, mg/dL	10.0	7.6	7.1	6.3	0.9	<0.01	0.45
VFA, mM							
Acetate	72.1	73.7	73.7	72.6	3.11	0.91	0.81
Propionate	22.8	23.4	22.9	23.4	1.00	0.69	0.94
Butyrate	14.5	14.1	14.0	13.5	0.90	0.35	0.93
Valerate	1.95	2.33	1.85	1.93	0.28	0.76	0.53
Branched- chain VFA	2.55	2.48	2.11	2.21	0.15	0.04	0.73
2-Methylbutyrate	1.24	1.16	1.04	1.06	0.09	0.09	0.69
Isobutyrate	0.90	0.86	0.81	0.84	0.03	0.09	0.33
Isovalerate	0.75	0.85	0.57	0.60	0.11	0.18	0.58
Acetate:Propionate	3.22	3.15	3.24	3.13	0.08	0.56	0.83

¹Covariate-adjusted least squares means.

²*P*-values of linear (L), and quadratic (Q) effects. Treatment $\times$ week interaction was not significant ($P > 0.05$) for all response variables.

and correlated with dietary digestible N (Spanghero and Kowalski, 1997, 2021). Unaccounted N represents actual N accretion in body tissues and accumulated error from sampling and measuring urine, feces, and milk N. A potential source of error in our trial was the analysis of N in dried feed and fecal samples. Recently, Morris et al. (2019) reported that about 5% of fecal N was lost during oven drying. Additionally, there is limited evidence that rates of urinary creatinine excretion are affected by dietary protein concentration, potentially affecting estimation of urinary N output when urine is spot sampled (Spek et al., 2013a). Alternatively, the change in BW for the duration of the entire study (reported by Barros et al., 2017) responded quadratically to dietary CP concentration with a maximum of 0.45 kg/d for the 14.4% CP treatment. In addition to differing in the pattern of response to dietary CP (linear vs. quadratic), unaccounted N and BW change also suggested different extents of N accretion. Considering estimates for the N content of gains in body reserve tissues (NASEM, 2021), unaccounted N suggested body tissue accretion was several times larger than that expected based on actual BW changes. Despite a likely overestimation of unaccounted N, our findings and those in Barros et al. (2017) suggested gains in BW and body N reserves at all levels of CP tested for late lactation cows (6 multiparous, 2 primiparous per pen). Based on changes in BW and BCS, Liu et al. (2021) estimated that both primiparous and multiparous late lactation cows accreted body protein when fed 13 or 16% CP (DM basis), but the lower CP condition depressed rates of BW gain, especially for primiparous

cows. As suggested by Reynolds et al. (2016), multi-lactation trials are needed to understand if the effects of lower-CP diets carry over into subsequent lactations, particularly for primiparous cows.

The lack of effect on NUE contrasted work with mid-lactation cows by Kalscheur et al. (2006), where CP spanned 12.3 to 17.1% of DM and Colmenero and Broderick (2006), where CP spanned 13.5 to 19.4% of DM. In these studies, losses in milk yield with reduced CP were less dramatic than those we observed (Kalscheur et al., 2006) or non-existent (Colmenero & Broderick, 2006). When testing 4 levels of dietary CP (13.6, 15.2, 16.7, 18.3% of DM) at 4 stages of lactation, Letelier et al. (2022b) found greater dietary CP and more advanced stage of lactation independently reduced NUE, but CP and stage of lactation did not interact to affect NUE. Although NUE was similar across treatments in our trial, we noted differences in urea pools (BUN, MUN, urinary urea-N) that have implications for the environmental reactivity of manure N (Burgos et al., 2007). The concentrations of urea-N in blood, milk, and urine decreased linearly with reducing dietary CP, as expected based on prior research (Spek et al., 2013c). As in previous reports (Broderick 2003; Colmenero and Broderick, 2006), the reduction in urinary N was mainly due to a reduction in urinary urea-N excretion, which decreased from 80.6% to 54.1% of the total urinary N when dietary CP concentration was reduced from 16.2 to 11.9%. Urea-N in plasma, milk, and urine results from hepatic detoxification of $\text{NH}_3\text{-N}$ absorbed from the rumen or produced by AA catabolism in the liver and peripheral tissues (Broderick and Clayton,

Table 5. Effect of decreasing dietary CP concentration on N partitioning for n = 16 pens of late-lactation dairy cows on wk 3, 7, and 11 and BUN on wk 4 and 12.¹

Item	Dietary CP Treatments (DM basis)				SEM	P-value ²	
	16.2	14.4	13.4	11.9		L	Trt*wk
Intake N, g/d	640	550	528	433	8.7	<0.01	0.03
BUN, mg/dL	13.4	10.2	8.8	5.5	0.35	<0.01	0.92
MUN, mg/dL	13.9	11.4	8.3	6.3	0.24	<0.01	<0.01
Milk N, g/d	177	161	152	127	5.9	<0.01	0.54
Fecal N, g/d	210	197	205	188	6.0	0.03	0.66
Urinary N, g/d	152	125	108	79	3.4	<0.01	<0.01
Urea N, g/d	122	93	74	43	2.8	<0.01	<0.01
Unaccounted N, g/d	102	67	66	38	5.3	<0.01	0.03
Percentage of N intake							
Milk N (NUE)	27.6	29.2	28.7	29.3	1.00	0.25	0.11
Fecal N	32.8	35.7	38.8	43.4	0.85	<0.01	0.24
Urinary N	23.8	22.9	20.6	18.3	0.72	0.23	0.01
Unaccounted N ³	15.9	12.1	12.1	9.0	0.97	<0.01	0.01
Urinary Urea-N, % of Urine N	80.6	73.9	68.2	54.1	1.21	<0.01	<0.01

¹Covariate-adjusted least squares means.

²Linear (L) effect of CP concentration level in the diet or interaction treatment by week (Trt*Wk); Quadratic effect was not significant for all response variables.

³Unaccounted N was calculated as the difference between N intake and the sum of N recovered in feces, urine, and milk.

1997; Nousiainen et al., 2004). Reynolds and Kristensen (2008) reported that the rates of urea re-entry to the gut increased and rates of excretion in urine decreased with lower dietary CP. Other studies have suggested a decline in MP efficiency and greater BUN at higher dietary CP levels, particularly when metabolizable AA were imbalanced (Haque et al., 2015). Therefore, a lesser concentration of N in urea pools can reflect greater efficiency of N capture into microbial protein and greater efficiency of AA use post-absorption.

Blood AA Concentrations

Tables 6 and 7 present the circulating concentrations of AA, which reflect a balance between the duodenal absorption of AA (supply), the removal of AA by the mammary gland, as well as AA metabolism by splanchnic and other tissues. Although MP supply was greatest for the 16.2% dietary CP treatment (Appendix Table A1), this treatment yielded the most milk true protein and fat, which may explain why the circulating concentrations of total AA (TAA) and of many individual AA were lesser at the 16.2% versus the 14.4% dietary CP treatment. In our trial, the overall serum concentrations of Arg, Ile, Lys, Thr, and Val were close to the averages reported by a meta-analysis by Patton et al. (2015); however, concentrations of His, Leu, Met, and Phe were lower. Indeed, plasma concentrations of His (19 $\mu\text{mol/L}$) and Met (16 $\mu\text{mol/L}$) in our trial approached the values observed in AA subtraction studies by Weekes et al. (2006; His = 10 $\mu\text{mol/L}$ and Met = 7 $\mu\text{mol/L}$; 9% CP basal diet) and Doelman et al. (2015; His = 13 $\mu\text{mol/L}$, Met = 10 $\mu\text{mol/L}$; 11.2% CP basal diet). Given the diets and production levels observed in the current experiment, NASEM (2021) predictions of the utilization efficiency for most EAA were greater

than target efficiencies at all dietary CP levels, with a substantial limitation apparently imposed by His supply (Appendix Table A2; Lapierre et al., 2020). As discussed by Lapierre et al. (2021) depletion of labile His pools including muscle carnosine, muscle anserine, and blood hemoglobin can contribute to maintaining His supply during deficiency, although the net contribution of these pools was probably insufficient to completely offset the severe deficiency imposed by the lower dietary CP level in our trial (Giallongo et al., 2017; Blancaert et al., 2021).

Despite the drastic linear reduction in dietary CP content, the serum concentrations of most individual EAA were similar across treatments in the present study. Although not significant, the serum concentration of His numerically declined as dietary CP content decreased ($P = 0.12$), especially at the 13.4 and 11.9% dietary CP levels. With decreasing dietary CP concentration, microbial protein often constitutes a larger fraction of MP and is lower in His compared with typical rumen-undegraded protein sources (Lee et al., 2012; Appendix Table A1). As an exception, serum Lys concentration responded quadratically to decreasing dietary CP similar to Raggio et al. (2004). Likewise, serum concentrations of NEAA were largely unaffected by treatment, apart from Asp, Cit, and Orn. Serum concentration of Cit decreased linearly as dietary CP decreased, whereas Asp and Orn concentrations responded in a quadratic fashion with concentrations maximal at moderate dietary CP. Previous literature has shown no differences in plasma Asp concentration with reductions in dietary CP concentration and has rarely reported Orn and Cit concentrations (Arriola Apelo et al., 2014; Cantalapiedra-Hijar et al., 2014; Giallongo et al., 2016). Omphalius et al. (2019) reported that a reduced MP treatment lowered plasma Cit and Orn, but tended to increase

Table 6. Effect of dietary CP concentration on serum AA concentration for $n = 16$ pens of late lactation cows on wk 4 and 12.¹

Item ³	Dietary CP% Treatments (DM basis)					<i>P</i> -value ²		
	16.2	14.4	13.4	11.9	SEM	L	Q	Wk
TAA	1899	2250	2152	2110	222	0.52	0.41	0.01
EAA	789	880	818	732	76	0.58	0.23	0.05
NEAA	1098	1378	1337	1377	148	0.21	0.48	0.02
BCAA	391	441	418	360	47	0.65	0.24	0.10
EAA:NEAA	0.73	0.64	0.64	0.55	0.03	<0.01	0.99	<0.01
BCAA%TAA	20.8	19.3	19.6	17.3	0.9	0.01	0.59	0.08
BCAA%EAA	49.3	49.4	50.7	48.7	1.7	0.93	0.55	0.99

¹Covariate-adjusted least squares means.

²*P*-values for linear (L), quadratic (Q), and week (Wk) effect of dietary CP. Treatment $\times$ week was not significant ($P > 0.05$) for all response variables.

³TAA = total AA concentration in the serum ($\mu\text{mol/L}$); EAA = essential AA ($\mu\text{mol/L}$); NEAA = non-essential AA ($\mu\text{mol/L}$); BCAA%TAA = branched-chain AA expressed as a percentage of TAA; BCAA%EAA = branched-chain AA expressed as a percentage of EAA.

Table 7. Effect of dietary CP concentration on individual serum AA concentration on wk 4 and 12 for n = 16 pens of late lactation cows.¹

Item	Dietary CP % Treatments (DM basis)					P-value ²		
	16.2	14.4	13.4	11.9	SEM	L	Q	Wk
EAA, $\mu\text{mol/L}$								
Arg	112.1	126.9	118.4	102.5	8.6	0.44	0.13	0.09
His	20.3	26.6	14.3	13.9	4.8	0.12	0.25	0.31
Ile	100.3	107.8	105.9	86.6	9.2	0.35	0.15	0.08
Leu	92.4	105.5	98.4	87.9	6.9	0.62	0.11	0.25
Lys	63.6	96.1	78.5	65.9	8.4	0.98	0.03	0.07
Met	14.7	16.4	15.9	15.1	1.1	0.81	0.28	0.31
Phe	28.5	30.7	30.3	30.4	1.8	0.48	0.62	0.01
Thr	99.9	112.0	103.4	96.4	12.1	0.77	0.50	<0.01
Trp	43.6	44.1	46.9	43.2	5.0	0.95	0.71	<0.01
Val	194.7	231	213	187.1	30.5	0.84	0.30	0.11
NEAA, $\mu\text{mol/L}$								
Ala	217.5	281	280.5	277.7	43.2	0.33	0.50	0.05
Asn	26.0	29.6	28.7	26.3	3.4	0.94	0.42	<0.01
Asp	5.4	7.8	7.1	5.9	0.86	0.66	0.05	0.02
Cit	53.8	49.5	50.2	40.4	4.23	0.01	0.32	0.43
Gln	259.3	301.1	277.5	290	17.0	0.28	0.47	0.56
Glu	67	93.1	87.2	82.8	9.7	0.29	0.19	0.95
Gly	227.7	315.1	317.8	358.5	64.6	0.17	0.78	<0.01
Orn	39.5	56.3	47.4	38.1	3.7	0.69	<0.01	0.16
Pro	78.3	81.5	91.2	84.2	12.1	0.61	0.72	0.14
Ser	87.4	119.1	111.2	119.9	14.9	0.53	0.12	<0.01
Tyr	39.4	41.7	42.5	42.0	3.9	0.60	0.76	0.21
3-MH ³	3.53	3.87	4.28	4.58	0.29	0.01	0.89	0.03

¹Covariate-adjusted least squares means.²P-values for linear (L), quadratic (Q); treatment $\times$ week (Trt $\times$ Wk) effect of dietary CP was not significant for all variables.³3-MH = serum 3-methyl-histidine concentration.

Asp concentrations compared with a higher-MP treatment. Consistent with previous work, with decreasing dietary CP, our results generally suggested decreasing ureagenesis was associated with lesser concentrations of the urea cycle metabolites Cit and Orn (Müller et al., 2021). However, in our study, lower serum concentrations of Orn in the 16.2% dietary CP treatment drove a quadratic response to dietary CP through unclear mechanisms. Our trial targeted similar dietary energy content across treatments but did not control metabolizable energy supply. Therefore, interactions in protein and energy metabolism may have influenced results. For example, Omphalius et al. (2020) found that abomasal infusions of glucose decreased plasma urea-N and tended to decrease Orn concentration, with no effect on plasma Cit concentration. We found serum and milk urea-N concentrations decreased linearly as dietary CP lowered. At lower dietary CP, a greater fraction of hepatic urea production may have been reabsorbed to the GIT, affecting serum and milk urea-N concentrations (Reynolds and Kristensen, 2008).

Although serum concentrations of TAA, EAA, NEAA, and branched-chain AA (BCAA) were similar across treatments in our trial, the ratio of EAA:NEAA and the percentage of BCAA in TAA decreased linearly

with decreasing dietary CP. In our trial, diets were predicted to reduce total EAA and total NEAA absorbed supplies proportionally as dietary CP decreased (1.27 to 1.30 g of NEAA per g EAA in MP). However, NASEM (2021) model predictions indicated that the individual EAA composition of MP differed slightly across treatments. When dietary CP decreased in our trial, the predicted percentage of MP from Met, Val, and Thr increased and His, Ile, Lys, Phe, and Trp decreased, where the maximum difference in the concentration of a given EAA was 0.1% of MP between treatments. As a result, the predicted EAA composition of MP may have more closely matched animal needs for lower compared with higher dietary CP treatments. Although this difference was quantitatively small, as pointed out by Lapierre et al. (2021), small differences in the metabolizable supply of a limiting AA can theoretically translate to large differences in milk protein synthesis and usage of other AA (e.g., 33 g of milk true protein contains 1 g Met, NASEM, 2021). The improved match of MP AA composition with animal demands may explain the reduction in EAA:NEAA at lower dietary CP concentration because a greater fraction of total EAA would be removed from circulation for use in the mammary gland and other tissues (Arriola Apelo et

al., 2014; Letelier et al., 2022a). Alternatively, lesser EAA:NEAA may stem from greater net mobilization of body protein reserves as dietary CP decreased, because NEAA such as Gly, Pro, and Ser are enriched in muscle and collagen (Ríus, 2019).

The decrease in BCAA%TAA with decreasing dietary CP was unexpected based on diet formulation. In contrast with other EAA, only a small fraction of BCAA are removed by the liver, and BCAA catabolism occurs primarily in extrahepatic tissues including the digestive tract and mammary gland (Lapierre et al., 2002). As a result, Martineau et al. (2019) showed that plasma concentrations of BCAA were more responsive to changes in MP compared with other EAA. This is because liver catabolism of BCAA is consistently minimal, whereas hepatic removal of other EAA increases with greater plasma concentrations (Hanigan, 2005). Therefore, a plausible explanation for the decrease in plasma BCAA%TAA we observed at lower dietary CP concentrations is that liver catabolism of non-BCAA reduced in response to lower dietary CP (retaining non-BCAA in circulation) whereas BCAA removal was less sensitive to dietary treatments, such that plasma BCAA concentrations were more reflective of dietary treatments. Omphalius et al. (2019) showed that feed restriction (decreased energy intake) decreased mammary plasma flow and AA uptake except for BCAA, and lesser MP independently increased mammary NEAA uptake, possibly indicating slowed rates of de novo NEAA synthesis and EAA catabolism by the mammary gland. This aligns with the association of decreased DMI with decreased serum BCAA as a percentage of TAA in our trial but disagrees with the decreased serum EAA:NEAA ratio we observed at lower dietary CP (and lower DMI). Potentially, the EAA composition of MP in our trial imposed substrate limitations or caused undesirable signaling effects that suppressed improvements in N efficiency at lower dietary CP levels (Doelman et al., 2015). Our trial assessed plasma AA concentrations and we did not intensively study AA metabolism by the mammary gland, skeletal muscle, or other tissues in response to lower dietary CP concentration. Additional research is needed—for example, to clarify the limits of postabsorptive metabolism to accommodate various metabolizable AA profiles at low dietary protein levels. Our trial illustrated that lower dietary CP can fail to improve N efficiency when it reduces DMI.

Blood 3-MH Concentrations

Table 7 presents the overall average of 3-MH concentration across wk 4 and 12. Serum concentrations of 3-MH increased linearly as dietary CP decreased, in-

dicating greater rates of skeletal muscle degradation at lower dietary CP (Akamatsu et al., 2007). The similar linear pattern observed in an analysis of a shorter term response of 3-MH to dietary treatments (Appendix Table A3) suggested that mobilization of these reserves had started within the first week of dietary CP restriction. Although our trial showed that lower dietary CP increased body protein degradation, net protein mobilization is the balance between protein degradation and synthesis rates (Sandri, 2013). Our trial did not directly measure protein synthesis, although BW and BW change were similar across treatments in our trial (Barros et al., 2017). Past work suggested greater rates of whole-body protein synthesis for cattle fed diets with higher CP concentration (Lobley, 2003; Cantalapiedra-Hijar et al., 2019) and suppressed synthesis of various tissue proteins in peri-parturient ruminants in a negative protein balance (Bell et al., 2000). In contrast, Raggio et al. (2007) found no differences in fractional synthesis rates of plasma proteins with varying levels of MP. Future long-term studies of reduced-CP diets may benefit from recent developments in techniques for estimating whole-body protein turnover (Cantalapiedra-Hijar et al., 2019), assessing gross changes in body reserves (McCabe and Boerman, 2020; Tebbe and Weiss, 2020), and characterizing the expression of mRNA and proteins related to major protein degradation and synthesis pathways (Nichols et al., 2017; Sadri et al., 2023). Late lactation is characterized by a decline in the number of active mammary epithelial cells, increasing reproductive demands, and alterations in endocrine status, potentially explaining why certain production responses to dietary CP concentration differ across stage of lactation (Letelier et al., 2022b). Recently, Piccioli-Cappeli et al. (2022) found that late lactation cows exhibited lesser whole body protein flux and a lesser somatotropin to insulin ratio compared with early-lactation cows. Therefore, future work is needed to understand differences in protein and AA metabolism as physiological processes shift in advancing lactation.

CONCLUSIONS

For the late-lactation Holstein cows in our 12-week trial, reducing dietary CP concentration had limited effects on ruminal concentrations of fermentation products, except for lowering $\text{NH}_3\text{-N}$ and BCVFA concentrations. Apparent digestibility of most nutrients was similar, except that linear and quadratic tendencies suggested that digestibility of pdNDF declined at CP levels below 14.4% of DM. With lower dietary CP, urinary urea-N output decreased but NUE was unchanged. In agreement with our hypothesis, serum con-

centrations of 3-MH suggested greater degradation of myofibrillar protein at lower dietary CP levels. Serum concentrations of TAA, EAA, NEAA, and BCAA were unaffected by dietary treatments. However, the ratio of EAA:NEAA lowered, which could indicate differences in protein metabolism induced by dietary CP level. In conclusion, although reducing dietary CP decreased excretion of environmentally-reactive N, more research is needed to design reduced-CP diets that promote desirable N partitioning (i.e., gains in body reserve protein, milk protein production) in late lactation.

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APPENDIX

Table A1. NASEM (2021) model predictions¹ of energy and protein related variables for the dietary conditions and cow performance observed in this study

Nutrient ²	Dietary CP Treatments (% DM)			
	16.2	14.4	13.4	11.9
NE _L , Mcal/d				
Supply	43.4	42.4	42.8	40.2
Requirement	41.1	40.1	38.7	34.3
Balance	2.3	2.3	4.1	5.9
MP, g/d				
Supply	2290	2043	1885	1613
Requirement	2345	2232	2140	1863
Balance	−55	−189	−255	−249
MP from Microbes	1320	1212	1144	1018
MP from RUP	970	832	741	596

¹Prediction based on NASEM (2021) model and the treatment means of DMI, milk yield, milk composition, BW, and BW change observed during the trial and available feed composition.

Table A2. Comparison of NASEM (2021) target efficiencies for metabolizable essential amino acid use with estimated efficiencies for the diet and animal conditions in our trial

AA	Target	Dietary CP (% DM)			
		16.2	14.4	13.4	11.9
Arg		0.54	0.6	0.66	0.71
His	0.75	0.85	0.92	0.97	1.00
Ile	0.71	0.64	0.68	0.70	0.70
Leu	0.73	0.75	0.79	0.82	0.82
Lys	0.72	0.72	0.77	0.79	0.80
Met	0.73	0.82	0.86	0.88	0.86
Phe	0.60	0.59	0.63	0.66	0.66
Thr	0.64	0.62	0.66	0.69	0.70
Trp	0.86	0.81	0.87	0.91	0.93
Val	0.74	0.71	0.75	0.77	0.77

¹Efficiencies were estimated for each treatment per NASEM (2021, p. 87) as the sum of metabolizable AA demanded for scurf, metabolic fecal protein, milk production, and growth, divided by available metabolizable AA for these functions after accounting for urinary endogenous and gestation net AA.

Table A3. Concentration of plasma 3-methylhistidine (3-MH) in the covariate period and in wk 1 of the experimental period measured in late-lactation cows fed different levels of dietary CP concentration¹

Plasma 3-MH, $\mu\text{mol/L}$	Dietary CP% (DM basis)				SEM	P-value ¹	
	16.2	14.4	13.4	11.9		L	Q
Covariate period, wk 3	2.48	2.38	2.63	2.75	0.20	NA	NA
Experimental period, wk 1	3.65	4.18	4.03	4.50	0.21	0.02	0.98

¹Blood was drawn from all cows from the coccygeal vessels into Vacutainer sodium heparin tubes (BD Vacutainer, Becton Dickinson, Franklin Lakes, NJ) at 1100 h (4 h after feeding) on d 15 of the covariate period and at 1100 h on d 7 of the experimental period. Plasma was obtained by centrifugation at 1000 *g* for 15 min at room temperature and stored at -20°C until further analysis. Samples were composited (by equal volume) by pen and sent to Heartland Laboratories (Ames, IA) for analysis of plasma 3-MH by gas chromatography-mass spectrometry (GC-MS; Rathmacher et al., 1992). NA indicates not applicable.